Doerun Temp Record Diagnostic Report

Date of Analysis: 2026-09-07

Source: Wunderground

Data Description: average temperature values aggregated daily.

Variable Analyzed: average temperature

Units: degrees fahernheit

Observation frequency: Daily

Observation count: 2329

Time Range: 2020-01-01 to 2026-08-31

1. Summary and Overview

1.1 Executive Summary

This report presents a structured diagnostic evaluation of temperature behavior using time-series analytical techniques. The dataset was processed and evaluated for trend structure, distributional behavior, autocorrelation, cyclic characteristics, and anomaly presence. Results indicate no statistically detectable linear trend at the 0.05 level, meaningful persistence near a lag of 1 day, and a possible recurring spectral period near 332.71 days. Distribution testing suggested the Normal distribution provided the strongest distributional fit.

1.2 Structural Summary Panel

Diagnostic Area	Structural Result
Sample Size	2329 observations
Trend	Limited evidence of linear trend
Distribution Shape	Moderate skew
Distribution Fit	No tested distribution provided an acceptable fit
Outliers	6 detected
Autocorrelation	Moderate autocorrelation near a lag of 1 day
Spectral Structure	Possible recurring pattern near 332.71 days
Rolling Variability Structure	Evidence of pronounced instability bursts

1.3 Overview

This report provides a structured diagnostic analysis of the selected time series, average temperature. The goal is to identify statistical structure, trend behavior, distribution characteristics, and possible readiness for deeper analysis.

1.4 Analyst Context Note

This analysis focuses on understanding the statistical structure of the dataset, including variability, trend behavior, distribution characteristics, and possible cyclic structure. Results should be interpreted as exploratory diagnostics intended to support further investigation rather than definitive causal conclusions.

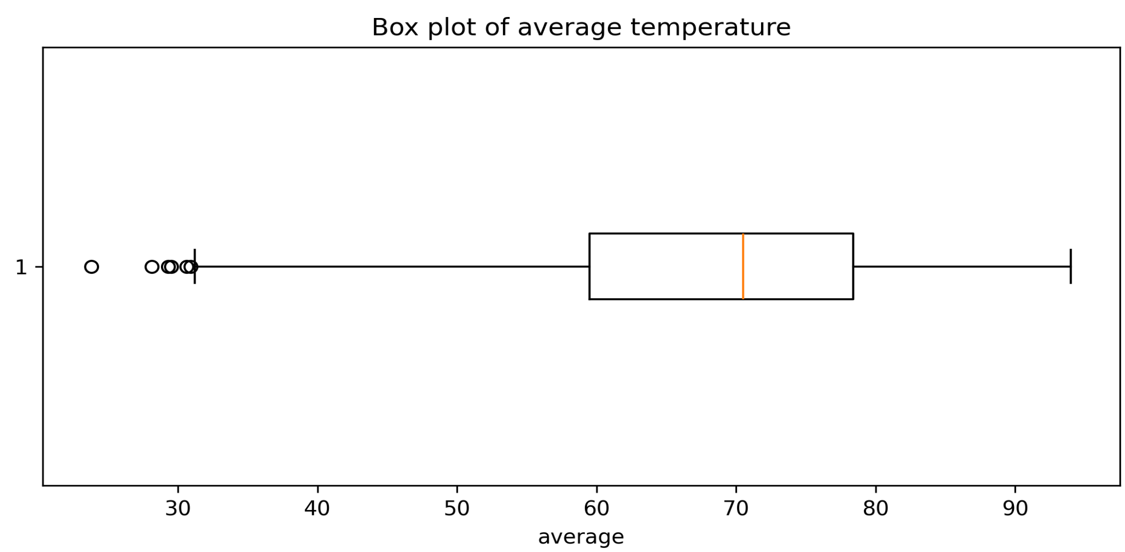
2. Summary Statistics

2.1 Summary Statistics table

Statistic	Value
Count	2329
Mean	68.03
Median	70.50
Std	12.52
Variance	156.79
Min	23.80
Q1	59.50
Q3	78.40
Max	94.00
Skew	-0.63
Kurtosis	-0.42
Lower Outlier Threshold	31.15
Upper Outlier Threshold	106.75
Number of Lower Bound Outliers	6
Number of Upper Bound Outliers	0

2.2 Box-and-Whisker plot

A box-and-whisker plot is shown below to visualize the distribution and outliers in the average temperature dataset. Note, multiple outliers may overlap visually in the box plot due to identical or near identical x-positioning.



3. Trend & Residual Analysis

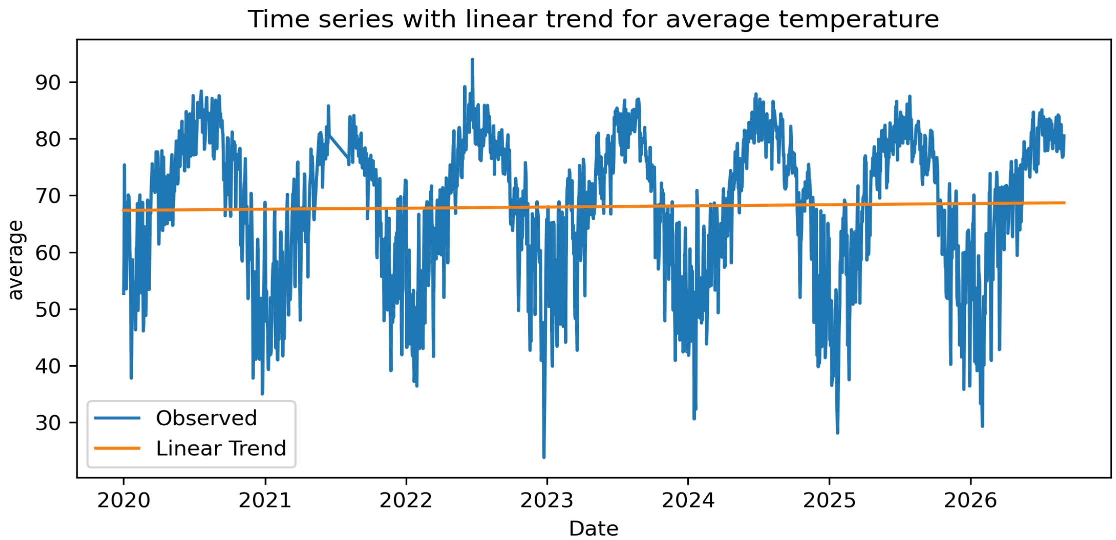
3.1 Linear Trend

Slope: 0.000569

R-squared: 0.0009

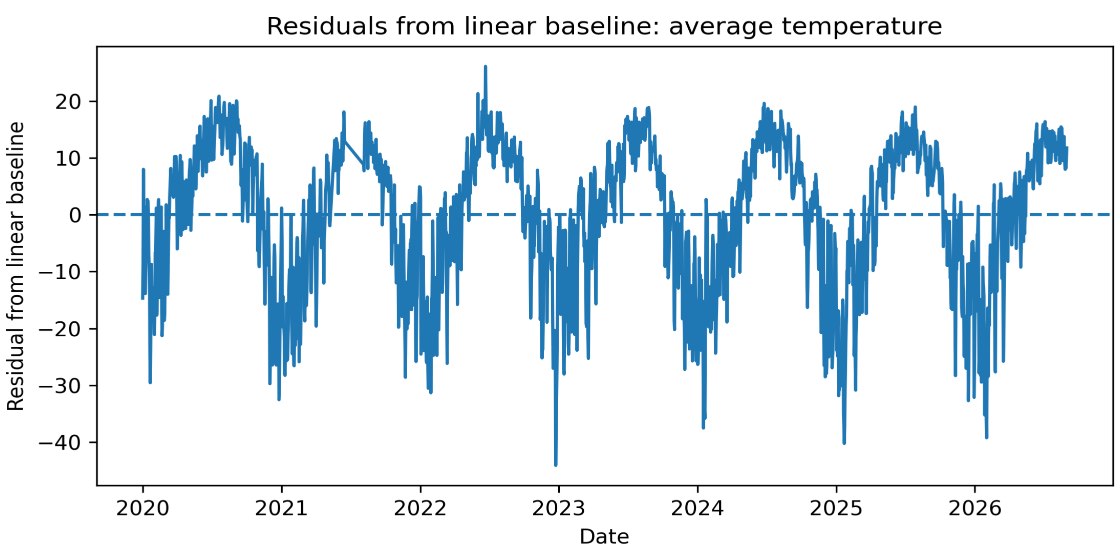
Trend p-value: 0.1407

No statistically significant linear trend was detected at the 0.05 level for the average temperature.



3.2 Linear Residuals

Because the fitted linear trend explains little of the variation, this plot should be interpreted as a baseline-deviation view.

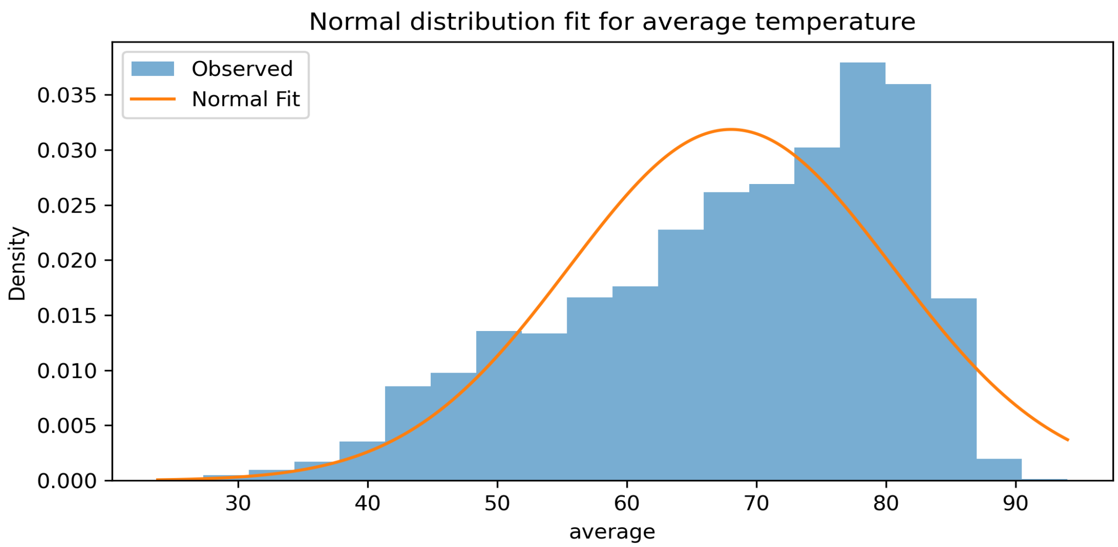


4. Distribution Comparison

4.1 Normal distribution fit

Metric	Value
Mean	68.0347
Standard Deviation	12.5191
KS Statistic	0.0876
KS p-value	0.0000

The KS test suggests the Normal distribution may not fully describe the dataset.

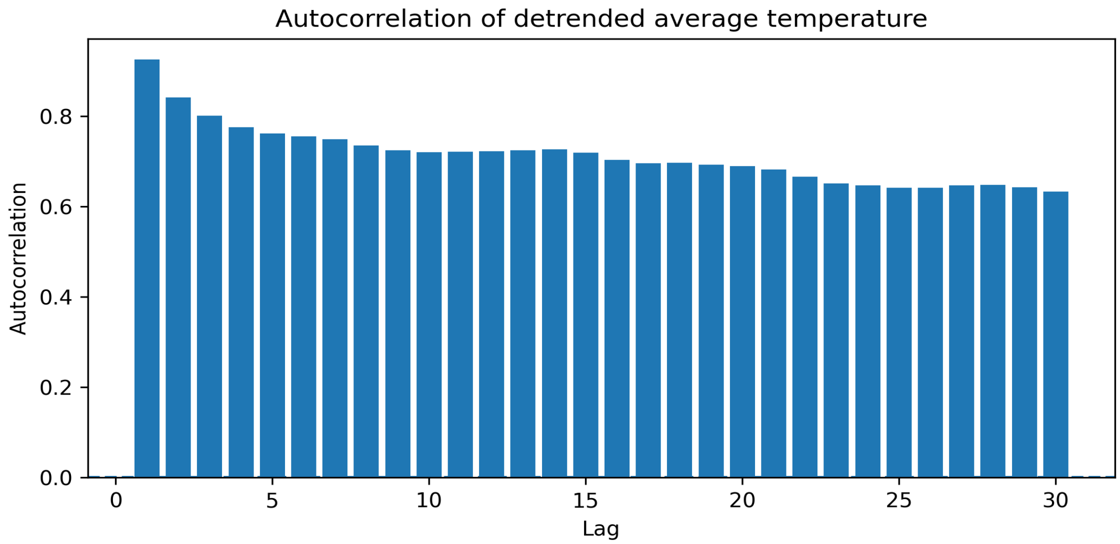


5. Lag and Periodicity

5.1 Autocorrelation Analysis

Autocorrelation measures whether later values in the dataset are similar to earlier values of itself.

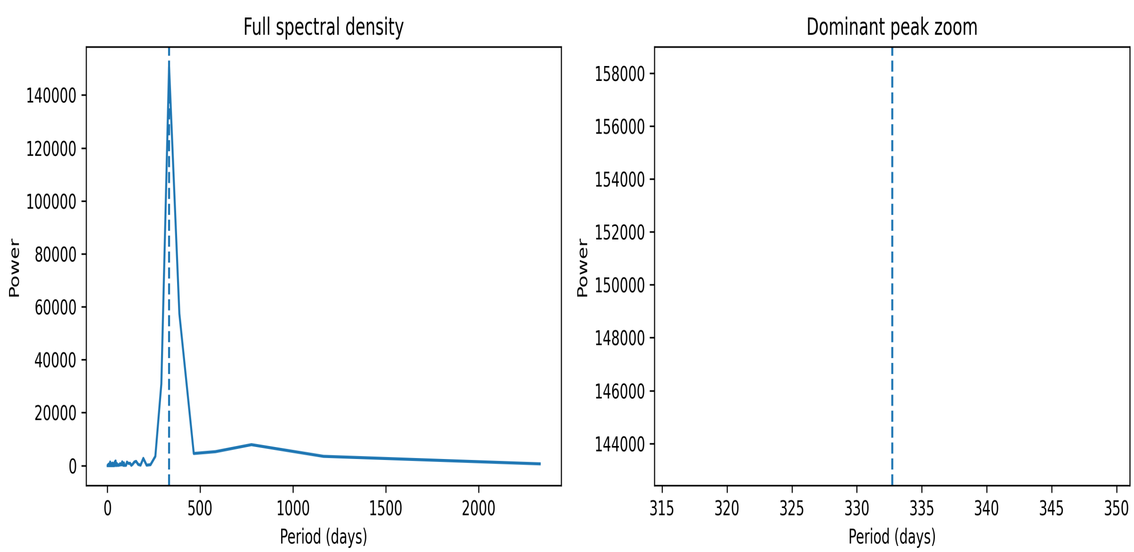
The strongest observed autocorrelation occurred near a lag of 1 day for the average temperature, with a correlation of 0.9250.



5.2 Spectral Analysis

Spectral analysis estimates which periods contribute the most power to the detrended time series. This helps identify possible recurring cycles.

The most dominant period noticed is approximately 332.71 days for the average temperature, based on the periodogram peak.



6. Local Instability & Threshold Concern

This section evaluates whether variability changes over time and whether the series crosses high-value thresholds often enough to suggest localized instability or elevated fluctuating periods.

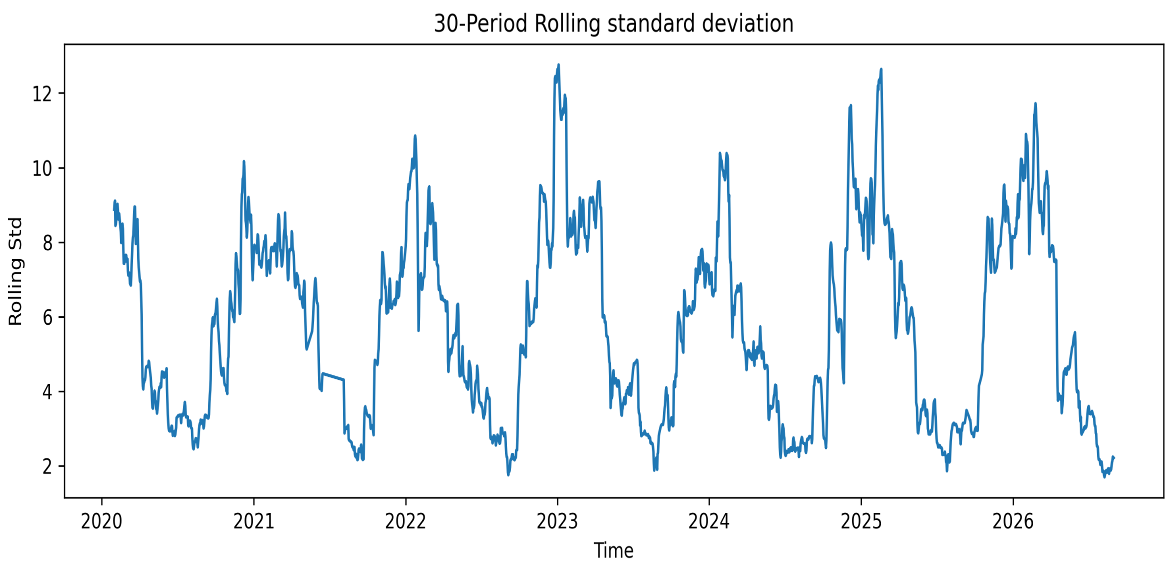
6.1 Rolling Variability

Rolling Standard Deviation Average: 5.75

Rolling Standard Deviation Average: 12.76

Rolling Variance Average: 39.56

Rolling Variance Maximum: 162.80



The average rolling standard deviation was 5.75, while the maximum rolling standard deviation reached 12.76.

The average rolling variance was 39.56, while the maximum rolling variance reached 162.80.

6.2 Threshold Exceedance

Threshold	Value	Count Above	Rate Above
90th Percentile	82.30	228	9.8%
95th Percentile	84.00	113	4.9%
99th Percentile	86.60	22	0.9%

Percentile exceedance rates provide a threshold-based view of extreme values. Higher exceedance counts indicate how often the series entered elevated-value zones relative to its own historical distribution. These thresholds do not define operational danger by themselves, but they provide a useful starting point for monitoring unusually high values.

6. Results

- 6 IQR-based outliers were observed under the IQR rule for average temperature.
- The analysis did not support a statistically significant linear trend for the average temperature.
- None of the tested distributions provided a statistically acceptable fit.
- Meaningful autocorrelation was noted around a lag of 1 day.
- A dominant period around 332.71 days was identified.
- There is pronounced instability around 2023-01-04 00:00:00.